Programming Project 4 Report

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Problem Statement:

The goal of the assignment was to create a 3D scene using OpenGL that displays several objects with textures. Each object uses a jpeg as a texture, so it is more realistic compared to plain colors. The program loads the texture images from files and maps them onto surfaces of the objects. The user can rotate the scene by using keyboard inputs so the objects can be viewed from different angles.

Design:

I started with a blank C++ file, but I used a lot of inspiration from the texture code examples on the class website. The main design focused on using a structure called Object that stores information about each object. This includes position, rotation, acceleration, size, and texture that is used. All the objects are stored in an array so they can drawn in a loop like previous projects. Textures are loaded by using a function called LoadTexture. This function reads an image file and converts the pixel values into a format OpenGL can use. Each object is then assigned a texture so that image appears on its surface.

Implementation:

Several textures were added to the project, some from the class website and some from screenshots of my grades, since I thought it would be funny. These images are stored in the textures folder and loaded when the program begins. The LoadTexture function reads each JPEG image and creates a texture that OpenGL can draw. Each object has its own texture ID which tells OpenGL which image to use. Then the objects are drawn, the correct texture is applied before the cube is rendered This allows each cube to display a different image. Keyboard controls allow the user to rotate the scene so the objects can be viewed from different angles. This makes it easier to see all the textures applied to each object.

AI Prompts:

Prompt 1: Why am I getting an undefined reference error when compiling my program?
Prompt 2: How do I link the libim and jpeg libraries in my MakeFile?
Prompt 3: Why is my program saying it cannot open my texture file?
Prompt 4: Why are some of my textures appearing black and white?
Prompt 5: How can I convert my image files so the texture loader can read them?
Prompt 6: What commands can I use to check the type of an image file in Linux?
Prompt 7: How can I add more custom textures to my OpenGL program?
Prompt 8: Give me a printout in C++ so the program controls are printed in the console.

Testing:
The program was tested by compiling and running the code several times. I checked that the OpenGL window opened correctly and that the objects appeared in the scene. I confirmed that each object showed the correct texture. I also tested the keyboard controls to make sure the camera rotates around the scene and zooms in/out. The program on my machine loads without any errors.

Figure 1: First Image of Scene

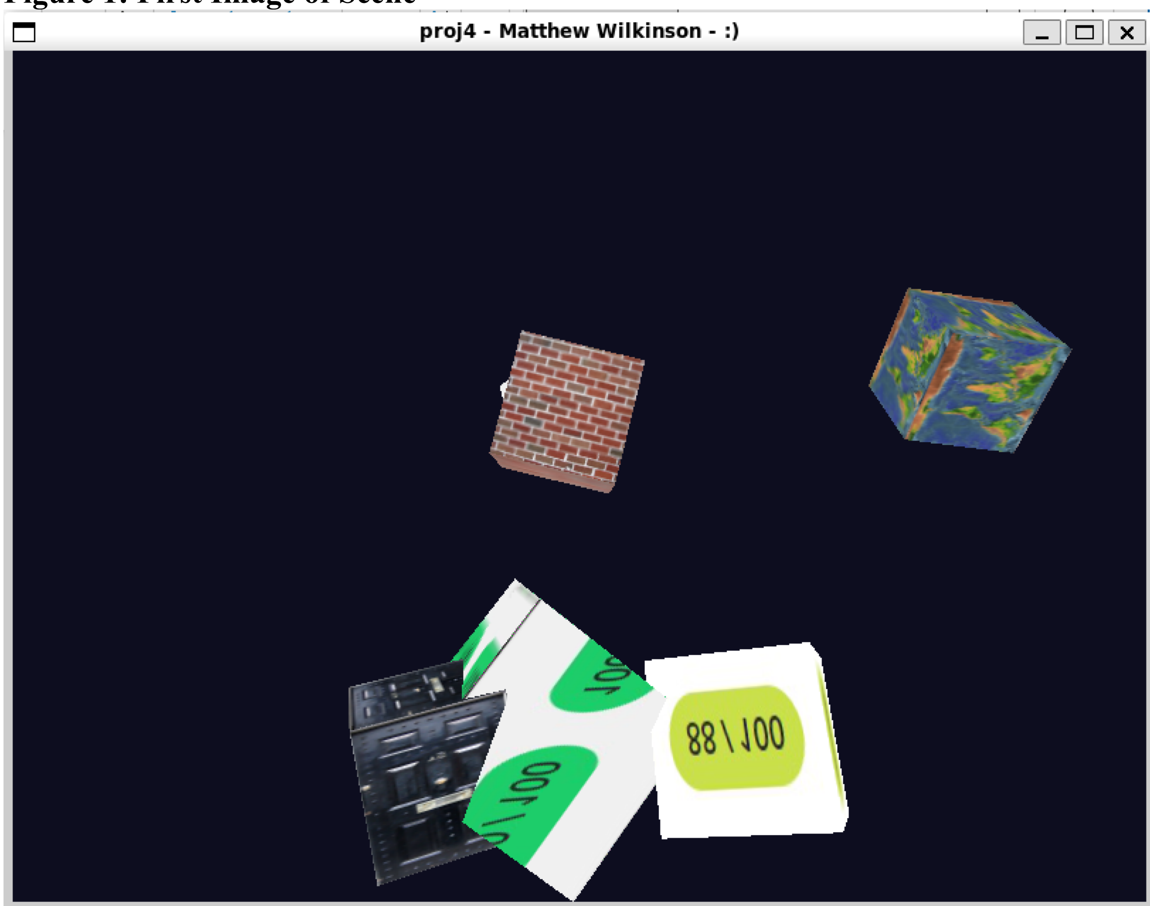

Figure 2: Second Image of Scene (ZOOMED OUT)

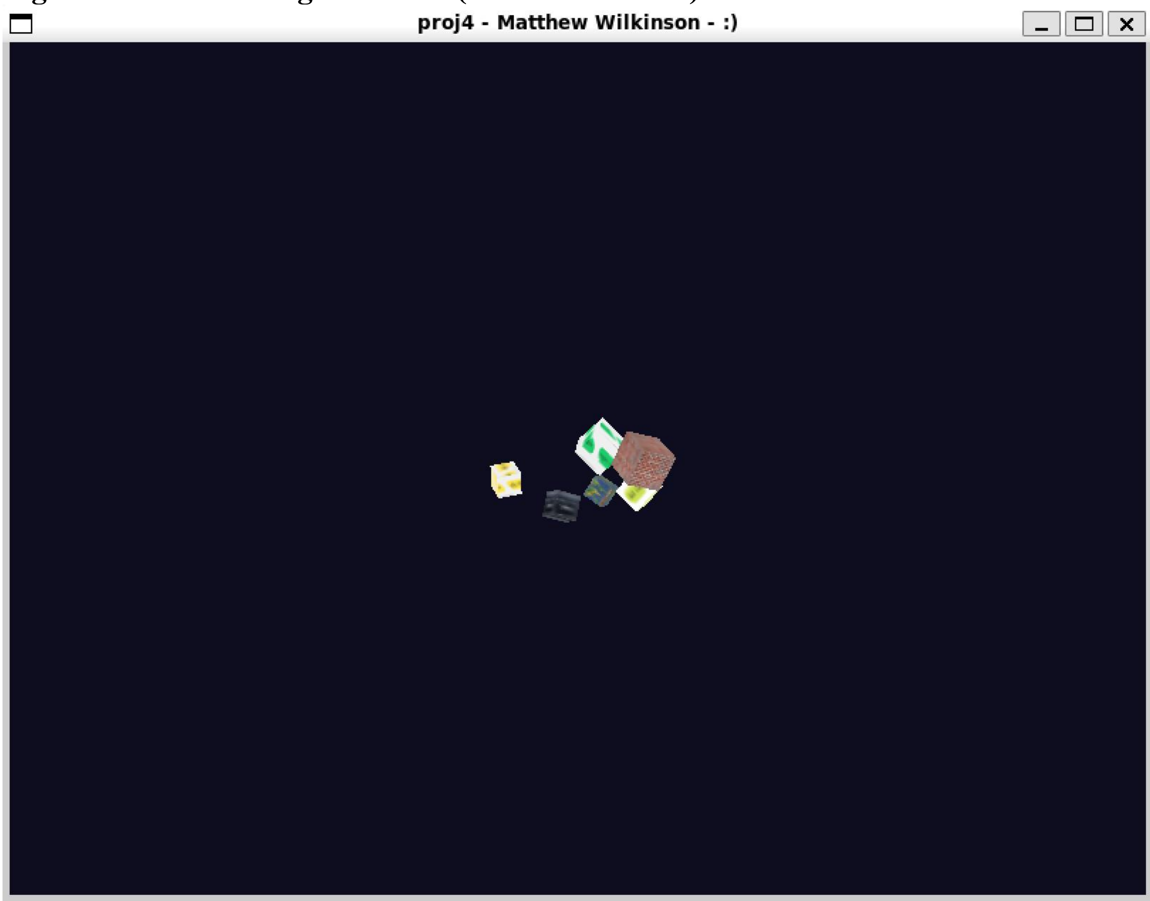

Figure 3: Third Image of Scene (ZOOMED IN)

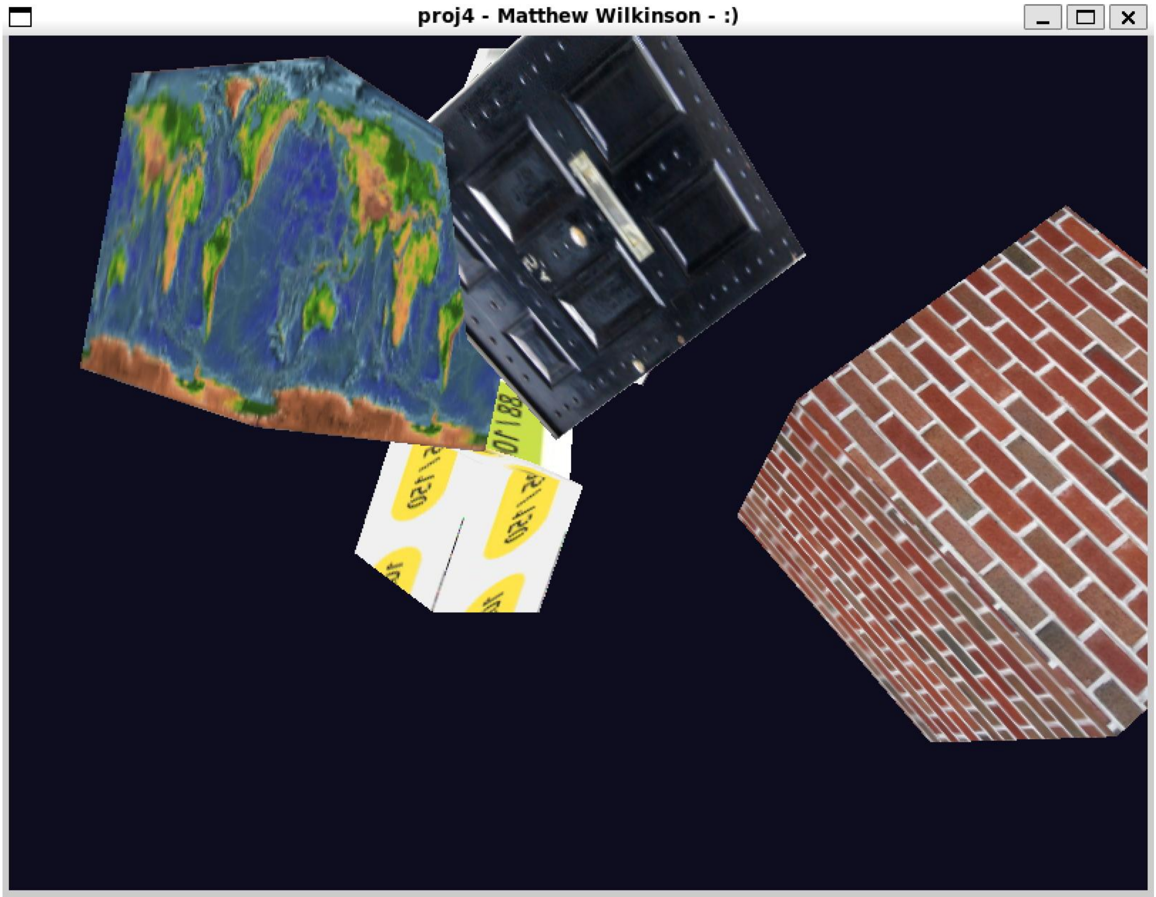

Figure 4: Terminal Screenshot

```
mmw071@BOOK-13EP48IKF6:/mnt/c/Users/mww61/git/graphics/graphics/proj4$ ./proj4

===== Controls =====
w : Rotate up
a : Rotate left
s : Rotate down
d : Rotate right
q : Zoom in
e : Zoom out
====================
```

Conclusions:

I believe the result of the assignment was successful, it produced an OpenGL window that visually appeared correct. The objects appear to display textures correctly. I might use a class for the Object if I did it again. It was interesting using Struct, I've just never really used it, and I thought it was interesting. It took me a couple of days, but it thankfully wasn't to bad thanks to the code provided on the class website.